CHAPTER 20

Nuclear Power Reactors

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We concentrate in this chapter on existing and some future power reactors and discuss the features common to each generic type as well as some peculiarities of special designs. Waste generation and handling at the power plants is also described generically.

Electric power producing reactors have been built to 1500 MW_e but all sizes in the range 400−1400 MW_e are common. With increasing size the cost of the electric power usually decreases. Power reactors of ~200 MW_e are competitive in some areas of high energy costs, such as in many developing countries. These smaller sized plants have some advantages: they are more reliable due to less stress of the components, they can be built more easily in remote areas because the smaller components are easier to transport, the electric grid may be more capable of accepting the moderate power addition, and they can more easily be designed with inherent safety features. Table 20.1 summarizes the characteristics of some operating nuclear power reactors.

In an attempt to systematise the continuous development of nuclear power reactors the concept of reactor generations has been introduced. In 2012 about 4 generations are discussed. The third has actually been divided into two different ones due to the clear trend break for the last generation where the novelty is not only the reactor but a coupled, optimised system of techniques.

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Table 20.1 Characteristics of some nuclear power reactors. Data refer to 2012

Type	Example	Power (MWe)	Fuel (% ^{235}U)	Can	Moderator	Coolant	Coolant temp. (°C)	Coolant press. (MPa)	Net eff. (%)	Burn-up (MWd kg^{-1})	Temp coeff.	Core size (h × d m^2)
PWR	Diablo Canyon 1, USA	1136	93 t UO_2 (4.4)	Za	H_2O	H_2O	320	15.5	32.1	45	Neg.	3.66 × 3.37
VVER*	Novovoronezh 5, Russia	1000	86 t UO_2 (4.4)	Za	H_2O	H_2O	324	15.7	32	<40	Neg.	3.53 × 3.16
BWR	Oskarshamn 3, Sweden	1450	126 t UO_2 (3.1)	Za	H_2O	H_2O	286	6.9	36.4	33	Neg.	3.68 × 4.60
PHWR	Bruce B5, Canada	915	135 t UO_2 (nat)	Za	D_2O	D_2O	305	9.2	30	7.7	Pos.	5.94 × 5.67
RBMK	Kursk 4, Russia	1000	218 t UO_2 (2.4)	ZN	C	H_2O	284	6.9	31	22	Pos.	7.0 × 11.8
GCR	Wylfa A1, UK	550	673 t U (nat)	Mgn	C	CO_2	414	2.7	26	3.5	Pos.	9.1 × 17.4
AGR	Hinkley Point B1, UK	655	119 t UO_2 (3.4)	St	C	CO_2	655	4.2	42	18	Pos.	8.31 × 9.11
LMFBR	Creys-Malville, France**	1240	31.5 t (Pu,U) $O_2 + UO_2$†	St	–	Na	545	0.1	41	70–100	Neg.	1.0 × 7.0

Za = Zircaloy, ZN = Zr + 1% Nb alloy, Mgn = Magnox, St = Stainless steel.
* PWR-like Russian design.
† 4.8 t Pu as 19% Pu MOX.
** Creys-Malvills is now permanently shutdown.

The basis for the division has been a development or improvement in certain features like e.g. safety, security and non proliferation, cost effectiveness, and the fuel cycle.

Generation 1: This generation is the early prototype reactors, typically designed in the 1950ies. Some examples are the Calder Hall-1 (1956–2003) in the UK and Shippingport (1957–1982) in the US. Many of these reactors were a "proof of principle" and the power levels were very low. The only remaining Gen 1 reactor operating today is the Wylfa Nuclear Power Station in Wales (UK) which was scheduled to be stopped in 2010, got an extension to late 2012, but is at the moment of writing not shut down.

Generation II. These reactors were typically designed in the 1960ies and were commercial reactors designed to be both reliable and economical. Most commercial power producing reactors operating today is of this generation. It include pressurized water reactors (PWR), CANada Deuterium Uranium reactors (CANDU), boiling water reactors (BWR), advanced gas-cooled reactors (AGR), Reaktor Bolshoy Moshchnosti Kanalniy or High Power Channel-type Reactor (RBMK) and Vodo-Vodyanoi Energetichesky or Water-Water Power Reactors (VVER). The design life time for this generation of reactors were typically 40 years. However, many of them have got a substantial prolongation of this time due to upgrades and replacement of critical parts.

Generation III. These reactors are essentially Gen II reactors which have evolved in some areas like thermal efficiency, fuel technology, modularized construction, and safety systems. In the latter case there has been some development to systems able to handle any event without operators input for several hours. The Gen III reactors comprise e.g. the Advanced Boiling Water Reactor (ABWR) by GE Nuclear Energy (first on line in Japan 1996), the advanced PWR (AP-600) (approved concept 1998) by Westinghouse and the Enhanced CANDU 6 by Atomic Energy of Canada Limited. These reactors have an expected life time of about 60 years.

Generation III+, This a further evolutionary step using essentially the same concepts as previously. The main feature is that more passive safety systems are included. They rely on gravity or natural convection to minimise the impact of abnormal events. Some examples of GEN III+ reactors are: VVER-1200/392M, Advanced CANDU Reactor (ACR-1000), AP1000 (based on the AP600) European Pressurized Reactor (EPR), and Economic Simplified Boiling Water Reactor (ESBWR) based on the ABWR. The GEN III+ reactors are also expected to achieve a higher fuel burnup than their evolutionary predecessors. This will reduce fuel consumption and thus waste production.

Generation IV, This generation is till rather undefined but it should be clear that now it is not only the reactor construction that is the issue but rather a whole concept aiming at superior usage of fuel by recycling the fuel. Thus increasing the energy utilisation and at the same time decreasing the long term radiotoxicity of the used fuel, hence achieving a considerable improvement of both safety and security. In many cases one hears about fast reactors being GEN IV reactors but this is not the case since the first power

producing fast reactors (EBR I) went critical in 1951. In the GEN IV system it is assumed that there is a clear integration of both thermal reactors as well as fast reactors, recycling plants for the fuel and fuel fabrication. Regarding the fast reactors some "novel" combinations of fuel and coolant are currently discussed. The proposed recycling plats could also be designed to avoid a pure plutonium steam for proliferation resistance using a separation process similar to the Group Actinide EXtraction (GANEX) process, see Chapter 21. However, with the development of centrifuge based uranium enrichment, the proliferation path using reactor produced plutonium will become less and less attractive for countries desiring nuclear weapons.

20.1. THERMAL REACTORS

The large majority of nuclear power reactors is based on fission by thermal neutrons and the discussion will begin with this class of reactors.

20.1.1. Pressurized water reactors

As mentioned in Chapter 19, the name implies that a pressurized water reactor is cooled by hot high pressure water, either H_2O (PWR, VVER) or D_2O (PHWR). In the PWR and VVER types the coolant is also used as moderator whereas a separate D_2O containing moderator tank is normally used in the PHWR type. These power reactor types have several things in common: primary − secondary coolant circuits separated by heat exchangers (*steam generators*), a *pressurizer* to adjust primary system pressure, and often *chemical shim control* for adjustment of the excess reactivity with fresh fuel.

The pressurized water reactor is generally preferred for propulsion purposes (military surface vessels and submarines), partly because it can react faster to changes in power demand than many other types of thermal reactors.

(a) *PWR*. The main components of a pressurized light water moderated and cooled reactor (PWR) station have been described in §19.1 and Figure 19.2.

The important features of the PWR core are shown in Figure 19.12 and described in §19.13.1. The design with a pressure vessel without facilities for fuel handling under pressure necessitates a yearly shutdown for fuel replacement. However, this is not considered a great inconvenience as a PWR anyhow has to be shutdown on a regular basis for safety inspection of pipes, welds, etc, and for routine maintenance. The need to shut down the reactor and open the pressure vessel in order to replace fuel also makes the PWR (as well as the VVER) resistant to concealed nuclear proliferation.

Leaking tubes in the steam generators have plagued many PWRs. The leaks are usually caused by corrosion. Such leaks are normally fixed by plugging the affected tube. However, too many plugged tubes leads to a reduction in permitted power